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Department of Computer Science

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S.Y.B.Sc.(Data Science) Sem-I

Subject: DS-203-MJ-P Lab Course on DS-201-MJ-T and DS-202-MJ-T

Data Structure-I Practical Assignment-1 Solution

Date: 20/06/2025

Topic: Array in Python

1. Write a Python program to create an integer array of 5 integers and display the array items. Access individual elements through indexes.

```
import array as arr
a=arr.array('i',[1,2,3,4,5])
print(a)
print(a[2])
```

2. Write a Python program to create float array of 5 integers. Append a New Item to the End of an Array.

```
import array as arr
a=arr.array('f',[10.2,20.4,30.6,40.8,50.1])
print(a)
a.append(70.2)
print(a)
```

3. Write a Python program to create Character array.

```
import array as arr
a=arr.array('u',['a','b','c','d','e'])
print(a)
```

4. Write a Python program to create integer array. Insert a New value at 2nd index.

```
import array as arr
a=arr.array('i',[1,2,3,4,5])
print(a)
a.insert(2,100)
print(a)
```

5. Write a Python program to create an integer array of 10 integers. Display length of array.

```
import array as arr
a=arr.array('i',[1,2,3,4,5])
print(len(a))
```

6. Write a python program to create a array of (10,20,30,20,40,50), remove the first occurrence of a specified element 20 from an array.(remove)

```
import array as arr
a=arr.array('i',[10,20,30,20,40,50])
print(a)
a.remove(20)
print(a)
```

7. Write a python program to create a array of (10,20,30,20,40,50) and delete last element from array.(pop)

```
import array as arr
a=arr.array('i',[10,20,30,20,40,50])
print(a)
a.pop()
print(a)
```

8. Write a python program to read two 1d array and print sum of it.

```
import numpy as np
a=np.array([10,20,30,40,50])
b=np.array([10,20,30,40,50])
```

```
result = []
for i in range(len(a)):
    result.append(a[i] + b[i])
print(result)
```

9. Write a python program to get the number of occurrences of a specified element in an array.

```
import array as arr
a=arr.array('i',[1,2,10,3,4,5,10])
cnt=0
for i in a:
    if i==10:
        cnt=cnt+1
print(cnt)
```

10. Write a python program to store all even numbers in new array from given data.

```
import array as arr
a=arr.array('i',[1,2,10,3,4,5,10])
b=arr.array('i')
cnt=0
for i in a:
    if i%2==0:
        b.append(i)
        cnt=cnt+1
print(cnt)
print(b)
```

11. Write a python program to create array using numpy and traverse it.

```
import numpy as np
a=np.array([10,20,30,40,50])
for i in a:
    print(i)
```

12. Write a Python program to create integer array and access element stored at index [1][1] and also access first row and print it.

```
import numpy as np
a=np.array([[1,2,3],[4,5,6]])
print(a)
print(a[1][1])
print(a[1])
```

13. Write a python program to create 2D array using numpy and traverse it.

```
import numpy as np
a=np.array([[10,20,30],[40,50,60]])
for i in a:
    for j in i:
        print(j, end = " ")
    print()
```